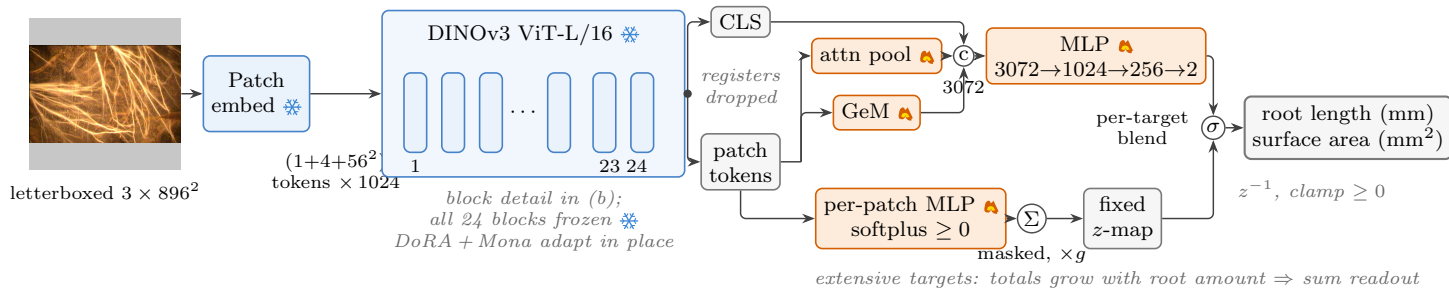
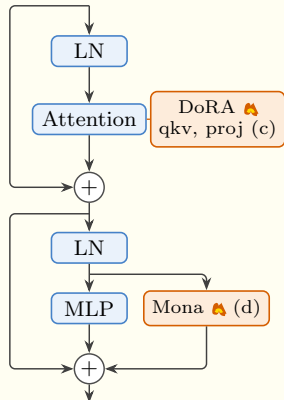
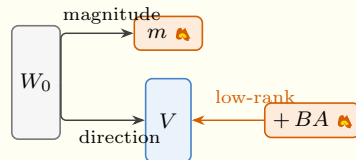


(a)

❄ frozen 🔥 trainable

(b) adapted ViT block ($\times 24$)

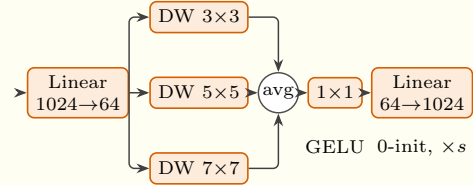
(c) DoRA, rank 32



$B=0$ at init $\Rightarrow$ training starts exactly from W_0

re-aims frozen attention toward root structures

(d) Mona adapter



reshaped to the patch grid; CLS & registers bypass

multi-scale convolution restores the locality a ViT lacks — thin roots are local texture

Trainable: 11.9 M / 315 M (3.78%) — DoRA 4.8 M · Mona 3.4 M · head 3.4 M · readout 0.3 M